

PUNIT MISHRA

Senior ML & AI Tech Lead — Turning Frontier AI into Shipping, Secure, Cost-Controlled Products

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EXECUTIVE SUMMARY

I lead a 13-person ML & AI team at SAP CX Data & AI — 11+ years at SAP, all told — where I turn hard AI and platform problems into products that ship in SAP's public and private cloud. I brought **SAP's own LLM (Nova)** into our natural-language-to-SQL stack as a first-class alternative to external frontier models — **raising SQL accuracy** while giving us model optionality and cost control. I architect **APEX**, our **Domain Data Access Agents** platform, and the security and multi-tenancy foundations that let us onboard enterprise customers safely and automatically, and I authored the ADRs the org builds against. I also work forward-deployed on our flagship customer, a Fortune 500 retailer, where my work cut answer latency more than half. My throughline: reduce dependence and risk, automate what used to be manual, and get real customer wins into production.

IMPACT HIGHLIGHTS

- | **NL-to-SQL accuracy:** bringing **SAP's own LLM (Nova)** into the stack with a date-arithmetic prompt fix moved decision-match **59.2% → 64.8%** and reranking **50% → 53%** — accuracy isolated and validated through a benchmark harness.
- | **Model optionality & cost control:** Nova as a first-class alternative to external frontier models reduces single-vendor dependence, with NL-to-SQL shipping in SAP's **2026 public and private cloud releases**.
- | **Customer-facing speed:** cut a Fortune 500 retailer's in-store concierge answer latency from **~25s to ~10s** and added grounded follow-ups that stop product drift — on SAP's first production enterprise customer.
- | **Closed a security risk class:** architected a hardened internal-auth contract (ADR 0021) across four Java + Python services — verified identity replacing trust-by-network-position.
- | **Automated onboarding:** turned **APEX**, our **Domain Data Access Agents** platform, into direct app tenancy (ADR 0020) so tenants provision themselves via BTP formations instead of manual setup — fixing a real cross-tenant leak in the process.
- | **Leadership:** lead a 13-person team and set the group's architecture (via ADRs), SDLC/engineering standards, and AI ethics & compliance as an **SAP Security Expert**.

EXPERIENCE

SAP

Feb 2014 – Present • 11+ yrs

Senior ML & AI Tech Lead | CX Data & AI

Sep 2018 – Present • San Ramon, CA

- | I brought **SAP's own LLM (Nova)** into our natural-language-to-SQL stack (S/4HANA + Ariba) as a first-class alternative to Claude — a self-hosted serving backend with OAuth2 and SAP-semantic prompts, an optional retriever-reranker, and seven benchmark lanes including SAP's first-ever Ariba coverage — **raising NL-to-SQL accuracy** while giving the business **model optionality, reduced dependence on external frontier models, and cost control**; the stack ships in SAP's 2026 public and private cloud releases.
- | I architect **APEX**, our **Domain Data Access Agents** platform, and converted it from manual, XSUAA-only tenancy into **direct app tenancy** (ADR 0020) where the platform owns its SaaS subscription lifecycle and tenants provision automatically via BTP formations — **automating self-service onboarding**, unifying identity, and fixing a real cross-tenant leak.
- | I architected and authored the **hardened internal-auth contract** (ADR 0021) — a shared secret validated by constant-time compare before any tenant header is trusted, canonical tenant resolution, and IAS identity alongside XSUAA — **closing a real cross-tenant escalation and data-leak risk** across four services, opt-in and backward-compatible so nothing broke.
- | As **forward-deployed engineer** on our flagship enterprise customer (a Fortune 500 retailer), I cut “ask a product” answer latency from **~25s to ~10s** and added grounded follow-ups that stop product drift — directly improving the in-store customer experience.
- | I modernized the legacy Shopping Assistant's vision into first-class **Domain Data Access Agent** services behind a unified **Agent Gateway** and an **MCP-based service Hub** — turning aging code into reusable, discoverable services.
- | I set technical direction for the group — guiding architecture through ADRs, owning AI ethics and compliance, introducing Ray, and standing up multi-model serving (self-hosted Nova plus Claude/Qwen/Granite-class) on bare-metal H100 clusters with automated eval harnesses.

Senior Software Engineer | Thunderbridge AI / SAP Sales Cloud

Feb 2014 – Sep 2018 • SF Bay Area

joined via CallidusCloud — acquired by SAP for \$2.4B

- | Built the Thunderbridge AI ML backend (Python/Dataiku, multi-tenant) and the CI/CD backbone; ported the full stack to GKE on-prem (Anthos) with Istio.

EARLIER

IBM — Automation Systems Engineer

Oct 2012 – Dec 2013

Built a data-validation automation framework and reusable Selenium/Java test APIs, adopted across the firm, for a major US bank.

CORE COMPETENCIES

AI/ML technical leadership (team of 13) • Cost control & vendor-dependence reduction • Enterprise security & compliance architecture • Multi-tenant SaaS at scale • LLM serving & evaluation (self-hosted + frontier) • Natural-language-to-SQL / applied LLM systems • Architecture governance via ADRs • Forward-deployed customer delivery • Identity & auth (IAS, XSUAA, OAuth2) • MCP-native agent services & gateways • Python & Rust engineering • SDLC & engineering standards

EDUCATION

UNIVERSITY OF CALIFORNIA, BERKELEY — B.S. Electrical Engineering & Computer Science

2012